

Ngoc (Ruby) Ngo

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EDUCATION

New York University

Master's in Data Science

New York, NY

Expected Graduation May 2027

- Relevant coursework: Database Systems (A), Computer Vision, GenAI Accountability (A), Data Visualization for ML

Dickinson College

Bachelor of Arts in Quantitative Economics & Music-Performance Emphasis

Carlisle, PA

Sept. 2018 – May 2022

- GPA: 3.92, Honors: Summa Cum Laude, Phi Beta Kappa, Dean's List

TECHNICAL SKILLS

Programming Languages: Python (PyTorch, Pandas, scikit-learn), R, SQL, Shell Scripting - Linux

Quantitative Skills: Deep Learning, Machine Learning, Econometrics, Optimization, Statistical Modeling

Software: Kafka, Snowflake, Hadoop, Spark, RStudio, Tableau, Jira, Jenkins, React, Bitbucket, Git, HPC, Stibo Systems, Qlik

WORK EXPERIENCE

NYU Langone Radiology Imaging | nnU-Net, PyTorch, HPC/Slurm

New York, NY

Machine Learning Intern

Jun. 2026 – Aug. 2026

- Engineered an **open-source deep learning segmentation pipeline** using **nnU-Net** and **PyTorch** to map microglial density across whole Macaque brain volumes, cutting processing time from weeks of manual labeling to hours.
- Architected an **incremental learning** strategy to eliminate spatial tiling artifacts, retraining the network on high-confidence inferences and elevating Mean Dice Similarity Coefficient from **75%** to **96.11%** on unseen validation slices

Bank of America | Linux, Python, Splunk, Oracle SQL, Ansible, Jira

New York, NY

Software Engineer II

Feb. 2024 – Jul. 2025

- Saved **\$2.5M** in potential FINRA fines by building a **Python automation** process that repaired **100M+** malformed **FIX trading records**
- Designed and implemented a more flexible and user-friendly repair execution process, reducing technical debt by **15%**
- Ensured stability and functionality of critical **RHEL** trading data processing systems by performing maintenance, implementing upgrades, and **leading the migration of legacy components** to modern technologies

Global Technology Analyst - Business Analyst

Jul. 2022 – Feb. 2024

- Analyzed **50M+** corrupt FIX messages post-trade to identify root causes and resolve systemic **data compliance** issues
- Developed **Python/Shell scripts** for **data reconciliation**; used comparative metrics to pinpoint discrepancies and cut manual correction volume by **40%**

TECHNICAL PROJECTS

Copyright Laundering | OpenRouter, Python, MCP, OpenClaw, Docker, GenAI Ethics

New York, NY

GenAI Accountability Data Scientist

Apr. 2026 – May 2026

- Developed a novel **red-teaming framework** to evaluate copyright laundering in **autonomous agents**, identifying an awareness-action gap where agents leveraged **multi-step tool workflows** to bypass standard safety guardrails
- Engineered a benchmarking suite using **MCP** and **OpenClaw** to evaluate agentic tool selection across 10+ tasks, quantifying compositional risk in autonomous systems

Cartier Jewelry Recommender | Supabase, Python, MongoDB, MySQL, React, Vercel

New York, NY

Data Scientist

Oct. 2025 – Dec. 2025

- Built an **end-to-end** personal stylist app leveraging outfit images and **in-database cluster metadata** to recommend across 500+ jewelry items
- Designed a **Supabase PostgreSQL** backend with **Edge Functions**, logging recommendation sessions across 25+ relational tables with low-latency queries

Google AI Hackathon - SnapEat | Python, Google Gemini, Azure, Django, Figma

Remote

Back-end Engineer

Apr. 2024

- Developed a **Generative AI-powered menu recommendation system** to support decision-making under user constraints, validated through **100+** test inputs to ensure consistent performance
- Built a Django backend that parsed menu photos with **Gemini** and ranked dishes against **user constraints** (dietary, budget), validated on **100+** test inputs and opportunities for improvement

LEADERSHIP & HONORS

Bank of America's Asian Leadership Network, Data Analytics Lead 2024 & 2025

New York, NY

New York Piano Society's 2024 Carnegie Hall Performing Member

New York, NY

Women in Economics Club, Vice President of Programming 2021 & 2022

Carlisle, PA

Scholar: Grace Hopper Celebration (2020, 2021, 2025, & 2026), Two Sigma Tapia (2020 & 2021)

Remote